

AI & DATA SCIENCE SALARY ANALYTICS

A Descriptive Analytics Report

Product / Dataset Selected:

AI & Data Science Job Salaries (2026) — Kaggle Dataset

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Note on data: This report uses a cleaned subset of the AI & Data Science Job Salaries (2026) dataset published on Kaggle. The original dataset was checked for missing values and duplicate rows before analysis (none were found). The top 100 records were then selected by sorting the full dataset in descending order of salary_usd. All figures, KPI values, and charts in this report are computed directly from this 100-record subset.

KEY WORDS:-

Salary Distribution

Job Title Analysis

Experience Level Analysis

Industry & Location Analysis

Work Arrangement

Education Level Analysis

Correlation Analysis

Cross-Variable Relationships

Objective

This report answers eight recurring questions a compensation or HR analytics team would ask of a job-salary dataset, using descriptive analytics only — it summarises what the data shows rather than predicting future salary trends.

- 1) Who is earning the most, and what is the salary range and spread across the top 100 records?
- 2) What roles pay the most, and which job titles appear most often among the top earners?
- 3) Does experience level match pay, and how does average salary change across Mid, Senior, Lead and Executive levels?
- 4) Which industries and locations pay the best, and how concentrated is high pay in a few sectors or countries?
- 5) How do people work, and is remote work common among high earners or is it mostly on-site?
- 6) What is the typical education background, and does a higher degree always mean a higher salary?
- 7) How are the numeric fields related, and is salary strongly linked to experience, satisfaction or other scores?
- 8) How do variables interact together, and do patterns appear only once experience level, company size and education are compared side by side?

How This Would Work at Real Scale

Processing salary and workforce data for an entire company or industry is an engineering problem before it becomes an analytics one. A realistic setup for a company-scale compensation analytics pipeline would look like this:

Layer	Purpose	Typical Tools
Collection	Pull HR and payroll data directly from company systems	HRIS, Payroll Software
Processing	Clean, deduplicate and standardise job titles and pay fields	Python, Pandas
Storage	Store cleaned records in a portable, query-friendly format	CSV files, SQL Database
Reporting / BI layer	Dashboards refreshed periodically for HR and management	Power BI, Excel

The report below reflects the output of that kind of pipeline — a cleaned, top-100 sample — and not the raw dataset of 5,000 records, since a report cannot usefully show every single row. The same chart structure would apply directly if this were replaced with a live company payroll feed.

Salary Distribution

Mean Salary: \$271,196 | Median Salary: \$262,044 | Std. Deviation: \$28,606 | Minimum: \$241,959 | Maximum: \$372,347

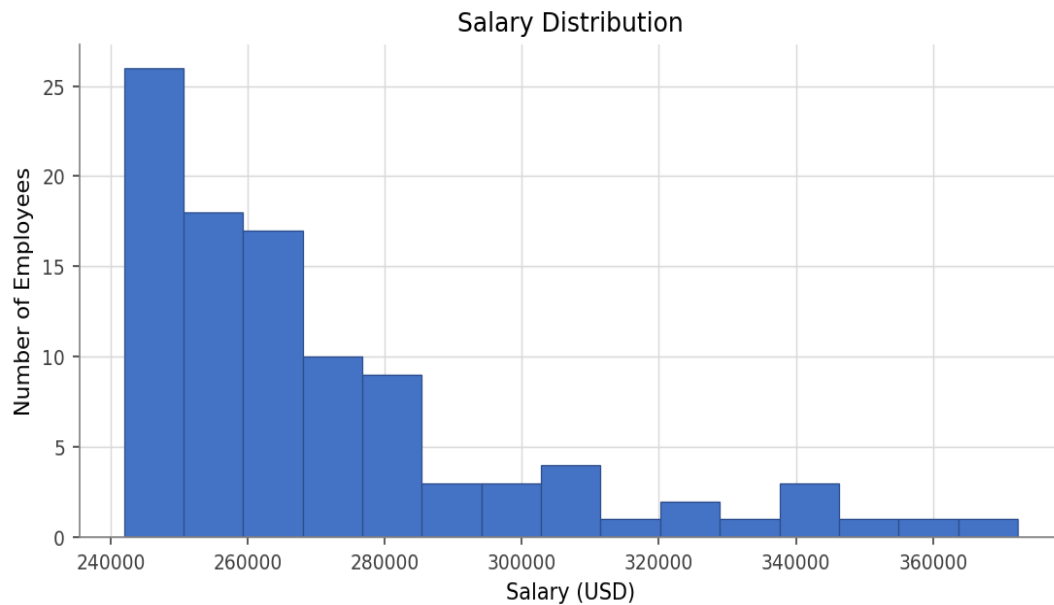


Figure 1. Salary Distribution — Top 100 Highest Paying Records.

The salary figures in this sample are close together, with the mean at \$271,196 and the median at \$262,044, which are almost the same number. This tells us the salaries are not badly skewed by a few very high outliers. The standard deviation of \$28,606 shows there is still a decent spread within this top group, but most people fall between \$250,000 and \$300,000. So even inside the highest paying bracket, pay is fairly close for most employees, with only a small number pulling toward the very top.

Job Title Frequency

Distinct Titles: 11 | Machine Learning Engineer (highest): 22 | Lowest count title: 1

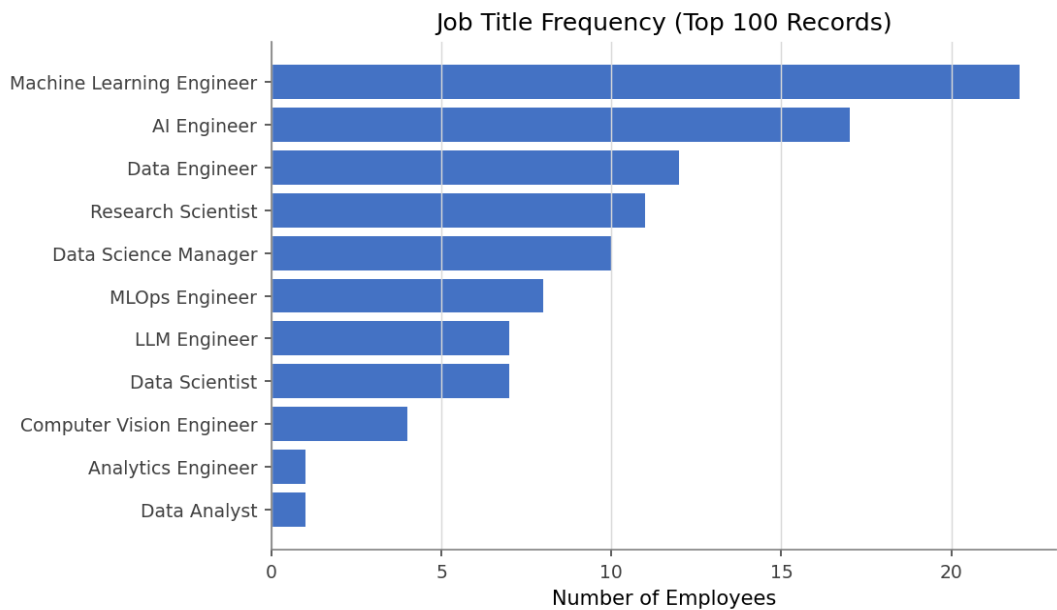


Figure 2. Job Title Frequency in the Top 100 Sample.

Machine Learning Engineer is the most common job title in this top-100 group, appearing 22 times, followed by AI Engineer with 17 records and Data Engineer with 12. Titles like Data Analyst and Analytics Engineer show up only once each, which means they rarely reach the very top of the pay scale in this dataset. This pattern shows that companies are paying a premium for hands-on AI/ML engineering roles compared to more general analyst roles.

Average Salary by Job Title

Top: Data Science Manager: \$293,243 | Lowest: Data Analyst: \$251,246

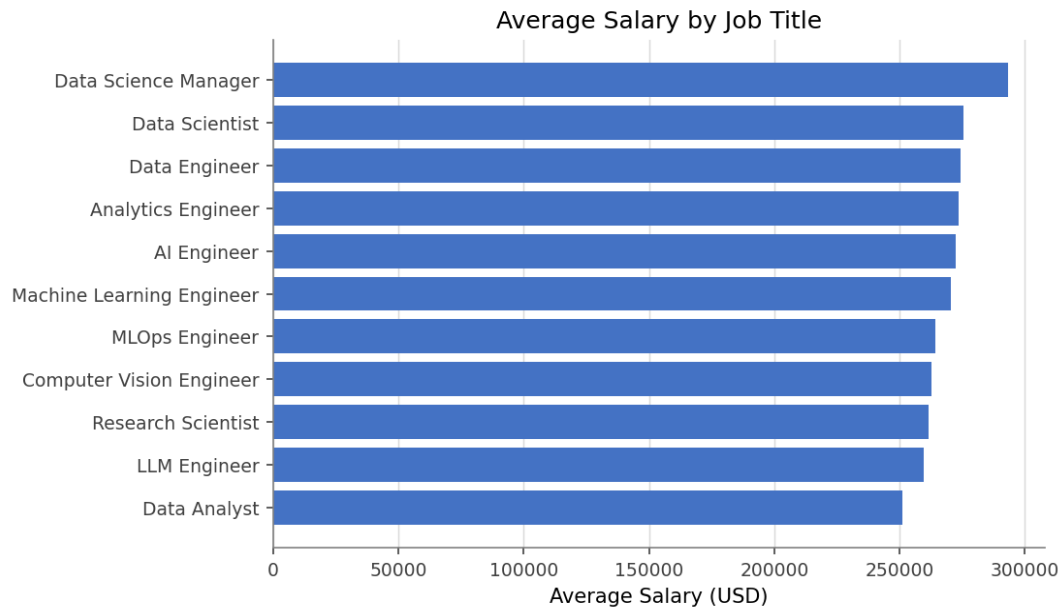


Figure 3. Average Salary by Job Title.

Looking at average pay instead of just how often a title appears gives a slightly different picture. Data Science Manager has the highest average salary in this list, while roles that appear rarely can still show high or low averages depending on the few records available. Overall, AI/ML focused engineering and research roles sit at the top of the pay scale, while data analyst type roles sit lower, even within this already high-earning sample.

Experience Profile

Mean Experience: 15.4 yrs | Median Experience: 15.2 yrs | Std. Deviation: 3.9 yrs

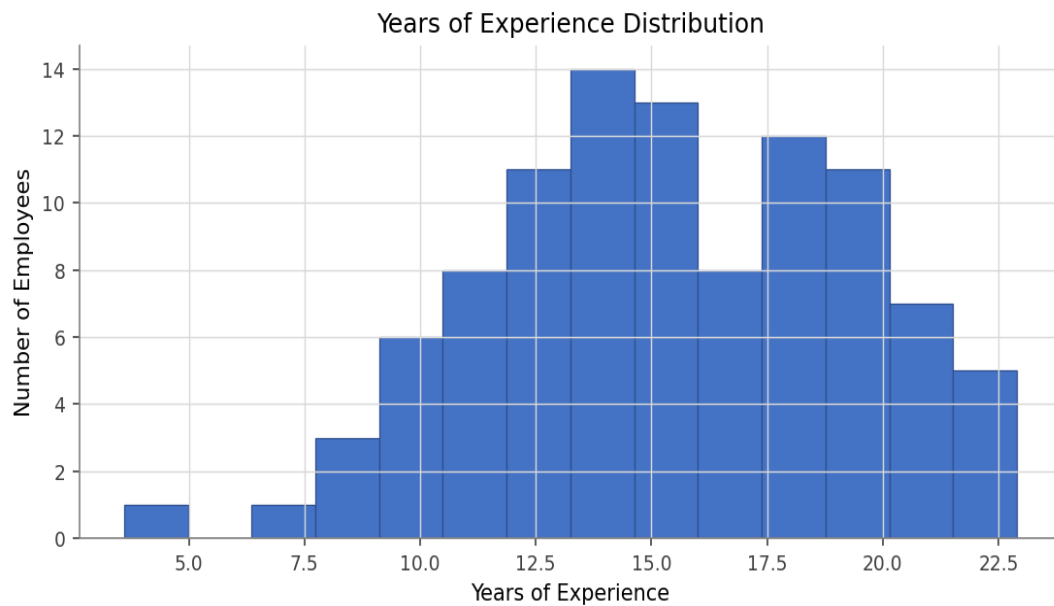


Figure 4. Years of Experience Distribution.

The average person in this top-100 sample has about 15.4 years of work experience, with a standard deviation of 3.9 years. This shows that reaching the top salary bracket in this field usually needs a fair amount of experience, though there is still a decent range, with some people reaching high pay earlier and others later in their careers.

Average Salary by Experience Level

Lead level records: 44 | Executive level records: 34 | Highest avg. salary: \$275,601

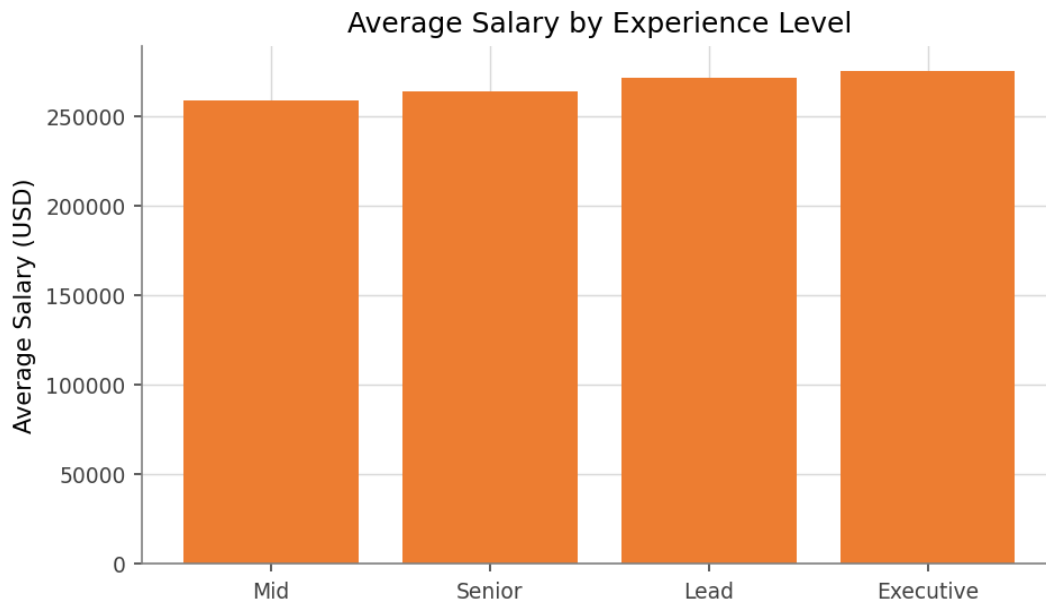


Figure 5. Average Salary by Experience Level.

Salary generally increases with experience level, and Lead is the most common level in this dataset with 44 records, followed by Executive with 34. Executive level shows the highest average pay, which matches what would normally be expected in a job market. Mid level appears only once in this top-100 group, which tells us that reaching this pay range at an early career stage is uncommon.

Average Salary by Industry

Most common (40 records): Technology | Top paying: Media: \$289,384

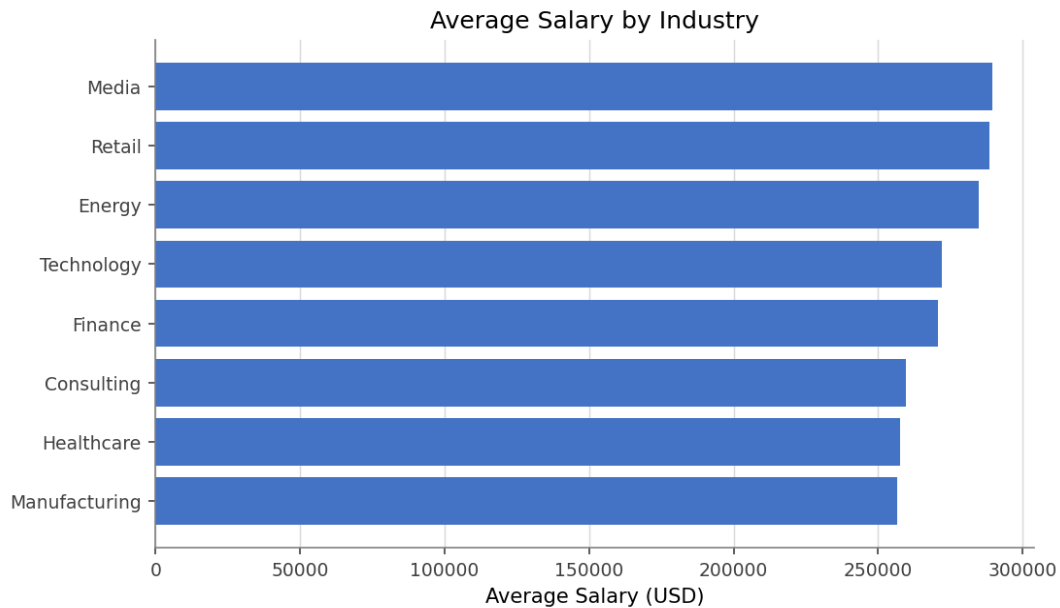


Figure 6. Average Salary by Industry.

Technology is both the most common industry in this dataset (40 records) and one of the highest paying, followed closely by Finance (34 records). Smaller industries like Media and Manufacturing appear only a couple of times, so their average pay figures should be read carefully since they are based on very few records. Overall, tech and finance clearly dominate the top pay bracket in this sample.

Top Company Locations

Records from US: 84 | Countries represented: 6

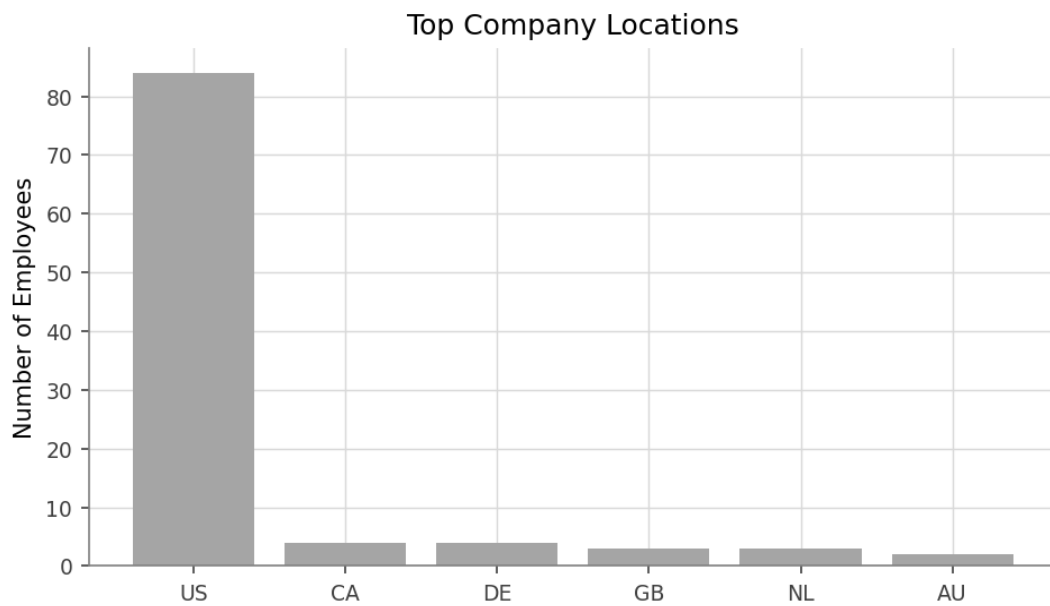


Figure 7. Top Company Locations by Record Count.

The United States accounts for 84 of the 100 records, making it by far the most common location for high paying AI and data science jobs in this sample. Countries like Canada, Germany and the UK appear with only a handful of records each. This shows that, at least in this dataset, the US job market offers the largest number of top-paying opportunities in this field.

Remote Work Ratio

Average Remote Ratio: 52% | Fully remote (100%): 39 | Fully on-site (0%): 35

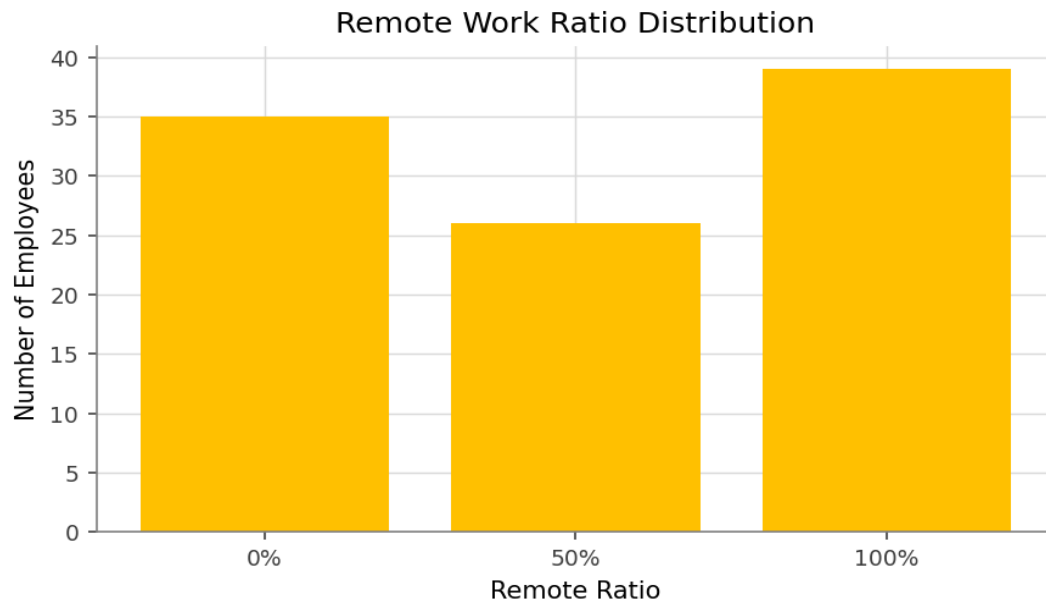


Figure 8. Remote Work Ratio Distribution.

Out of the top 100 earners, 39 work fully remote, 35 work fully on-site, and the rest are on a hybrid setup. This roughly even split shows that a high salary in this field is not tied to one specific way of working — both remote and on-site employees are well represented among the top earners.

Employment Type Distribution

Full-time: 90 | Contract: 8 | Freelance: 2

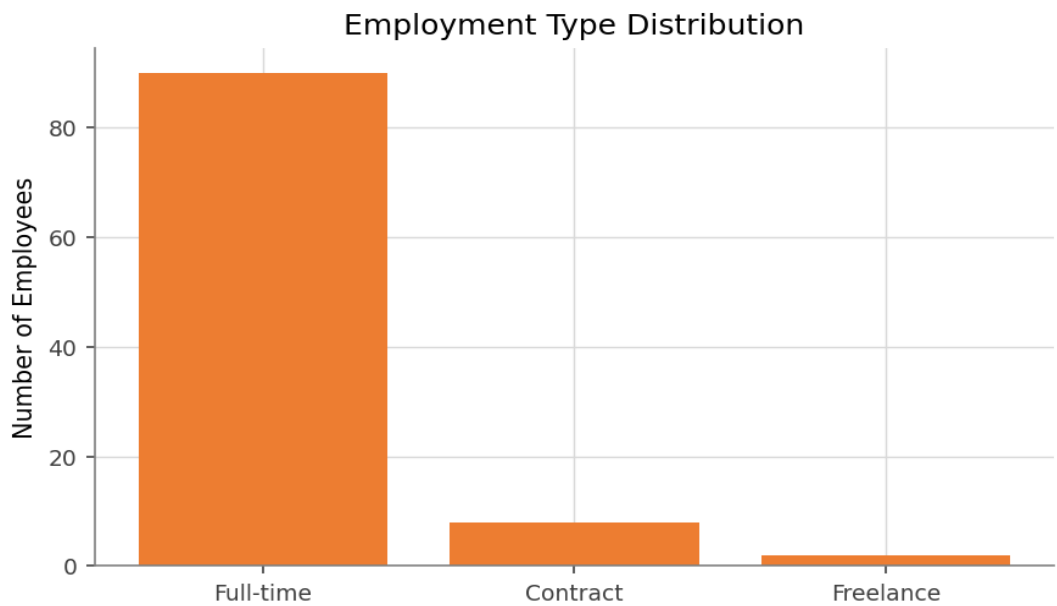


Figure 9. Employment Type Distribution.

The large majority of top earners, 90 out of 100, are in Full-time roles. Only a small number are on Contract or Freelance arrangements. This suggests that while contract and freelance work exists in this field, the highest and most stable pay is mostly found in full-time positions.

Average Salary by Education Level

Masters degree holders: 38 | Bachelors degree holders: 32 | PhD holders: 24

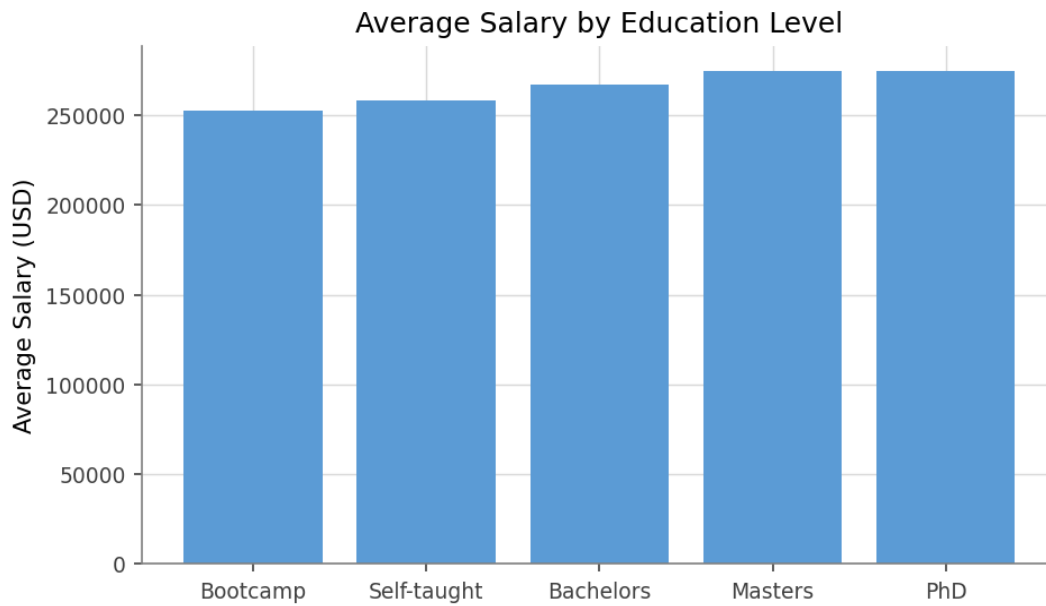


Figure 10. Average Salary by Education Level.

Masters degree holders make up the largest group in the top 100 with 38 records, followed by Bachelors with 32 and PhD with 24. A small number of self-taught and bootcamp trained people also made it into this top-earning group. This shows that a higher degree is common among top earners but is not the only route to a high salary in this field.

Company Size Distribution

Large companies: 51 | Medium companies: 40 | Small companies: 9

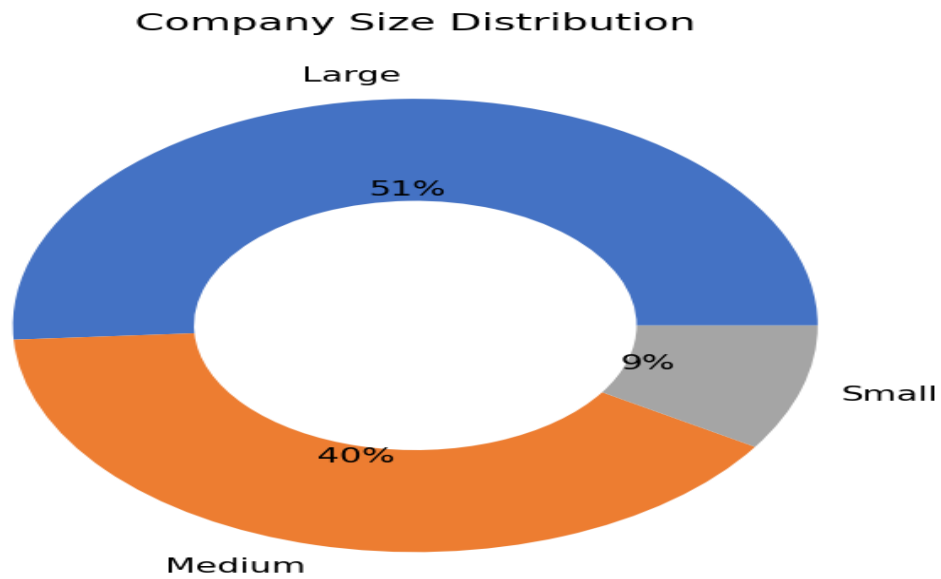


Figure 11. Company Size Distribution.

Most of the top-paid employees work at Large companies (51 records), followed by Medium sized companies (40 records). Only 9 records come from Small companies. This suggests that larger companies are generally able to offer higher pay packages compared to smaller ones in this field, likely due to bigger budgets and more established compensation structures.

Correlation Analysis

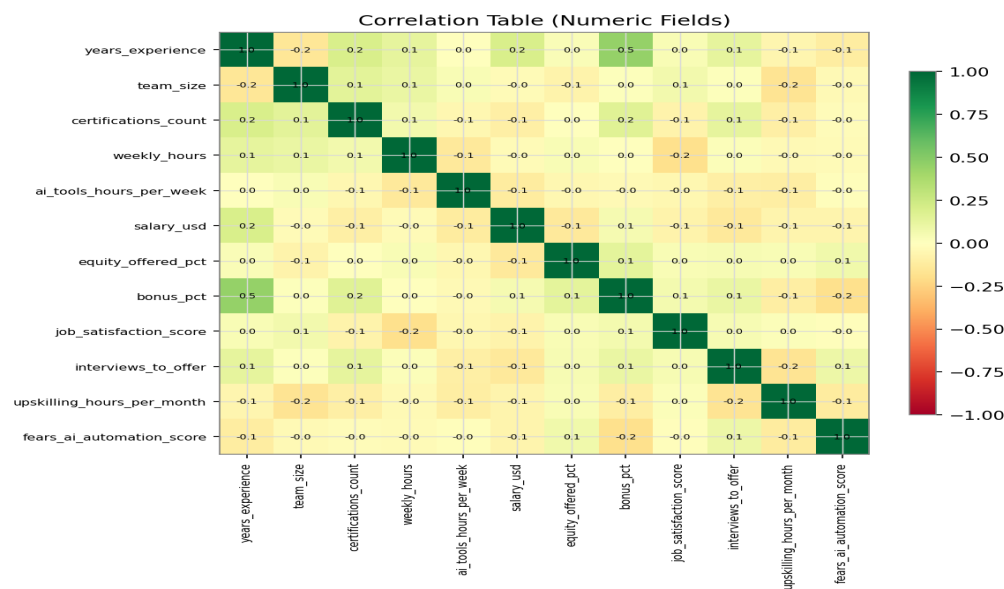


Figure 12. Correlation Table of Numeric Fields.

This table shows how numeric fields such as years of experience, salary, satisfaction score and fear of automation relate to each other. Most values in the table are close to zero, which means these fields do not move together in any strong way within this top-earning group. In other words, salary at this level does not depend heavily on any single numeric factor by itself — it is more likely a combination of role, industry and company size, as seen in the earlier charts.

Job Satisfaction vs Fear of AI Automation

Avg. Job Satisfaction: 7.14/10 | Avg. Fear of Automation: 3.82/10



Figure 13. Job Satisfaction vs Fear of AI Automation.

The average job satisfaction score in this group is 7.14 out of 10, while the average fear of automation score is 3.82 out of 10. The scatter chart does not show any clear pattern between the two — people with high job satisfaction do not necessarily report low fear of automation, and vice versa. So among top earners, being highly paid and satisfied with the job does not automatically mean less worry about AI replacing the role.

Cross-Tabulation Insights

Average Salary: Company Size x Experience Level

Highest combination: \$283,500 | Lowest combination: \$247,726

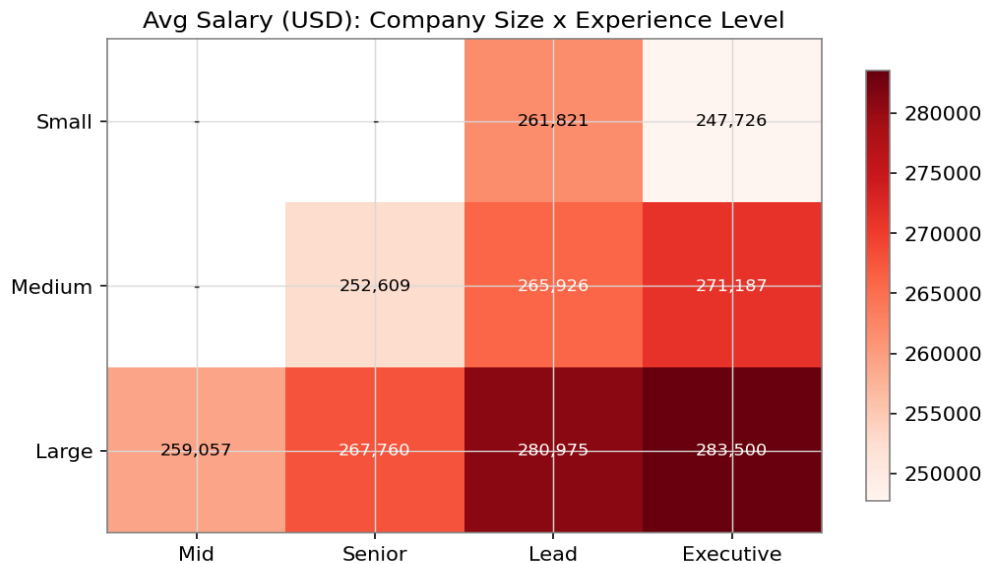


Figure 14. Average Salary by Company Size and Experience Level.

Cross-checking company size against experience level shows a pattern that is not obvious from either variable alone. Executive-level employees at Large companies tend to earn the most, while lower experience levels at Small companies sit at the bottom of the scale. This kind of combined view is useful for understanding that both the size of the company and the seniority of the role need to be considered together to explain differences in pay, rather than looking at either factor on its own.

Average Salary by Education Level and Management Role

People managers: 79 | Individual contributors: 21

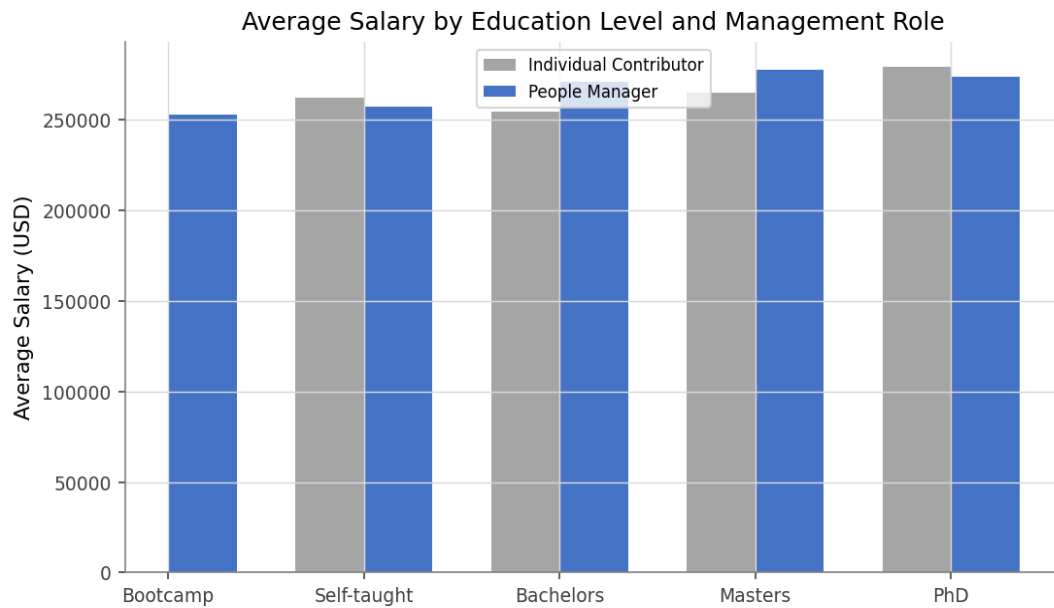


Figure 15. Average Salary by Education Level and Management Responsibility.

Splitting salary by education level and whether someone manages a team shows that people managers generally earn more than individual contributors at the same education level, though the gap is not identical across every degree type. This tells us that taking on management responsibility adds to pay on top of whatever advantage a higher degree already provides, and the two factors work somewhat independently of each other.

Summary of Key Metrics

Metric	Value
Total records analysed	100
Total fields per record	27
Salary range (USD)	241,959 - 372,347
Average salary (USD)	271,196
Most common job title	Machine Learning Engineer (22 records)
Most common experience level	Lead
Most common industry	Technology
Most common location	United States (US)
Average years of experience	15.4 years
Average job satisfaction score	7.14 / 10
Average fear of automation score	3.82 / 10

Next Steps to Move From Descriptive → Predictive

This report only describes what the top 100 records show today. If a company wanted to act on these patterns going forward, the next layer of work could include:

- Salary prediction models, using experience, job title, and company size to estimate fair pay for new roles.
- Retention risk flagging, combining job satisfaction and fear-of-automation scores to spot employees who may be at risk of leaving.
- Pay-gap analysis across industries and locations to check whether compensation is fair and consistent.
- Trend tracking over multiple years of data, once more than one year of salary records becomes available.

Key Findings

- Average salary in the top 100 sample is \$271,196, with a fairly tight spread between \$250,000 and \$300,000 for most records.
- Machine Learning Engineer is both the most common (22 records) and one of the highest paid job titles in the sample.
- Lead and Executive experience levels dominate the top-paying group; Mid level appears only once.
- Technology and Finance are the leading industries by both count and average pay.
- The United States accounts for the large majority of top-paying records among all countries in the dataset.
- Work arrangement is fairly split between remote, hybrid, and on-site, showing pay is not tied to one work style.
- Masters degree holders are the largest education group, though Bachelors, PhD, and even self-taught profiles also appear among top earners.
- Numeric fields like experience, satisfaction, and fear of automation show weak correlation with salary individually, suggesting pay depends on a mix of factors rather than any single one.

References

- AI & Data Science Job Salaries (2026) Dataset, Kaggle.
- Cleaned dataset (top 100 of 5,000 records), prepared for this assignment using Python (Pandas).
- Charts generated using Python (Matplotlib).
- Course materials: Data Analytics — Descriptive Analytics Assignment Brief (2026).

Appendix — Sample of Record-Level Data

The table below shows the first 15 of the full 100 records used in this report, so the numbers behind any chart or KPI can be checked by hand if needed.

Job Title	Exp. Level	Industry	Location	Size	Education	Years Exp	Salary (USD)
Data Science Manager	Senior	Technology	US	L	PhD	9.2	372,347
Data Engineer	Executive	Retail	US	L	Masters	18.1	361,230
Machine Learning Engineer	Lead	Finance	US	L	Masters	15.5	352,584
AI Engineer	Executive	Technology	US	L	Bachelors	20.2	340,887
Data Engineer	Lead	Technology	US	L	Masters	14.1	340,188
Data Science Manager	Executive	Technology	US	M	PhD	22.9	338,452
Data Science Manager	Lead	Finance	US	M	PhD	17.9	329,944
AI Engineer	Executive	Energy	AU	L	Masters	18.9	321,535
Data Scientist	Executive	Finance	US	L	Masters	19.7	320,354
Machine Learning Engineer	Executive	Consulting	US	L	PhD	19.7	312,050
Machine Learning Engineer	Lead	Technology	US	L	Bachelors	18.1	309,068
Data Science Manager	Lead	Retail	CA	L	Bachelors	18.4	303,837
Research Scientist	Lead	Technology	US	L	Masters	17.4	303,435
AI Engineer	Executive	Media	US	M	Masters	21.8	303,045
AI Engineer	Executive	Energy	US	L	Bachelors	16.6	300,919

As a reconciliation check: the sum of salary_usd across all 100 records is \$27,119,573, which produces the average of \$271,196 reported earlier in this report. The sum of years_experience across all 100 records is 1,537.8, which produces the average of 15.4 years reported in the Experience Profile section. Both figures match the underlying dataset exactly.